

Automatic vision-based calculation of excavator earthmoving productivity using zero-shot learning activity recognition

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ABSTRACT

Recently, vision-based methods have been widely used to analyze the construction productivity based on onsite videos owing to their low cost, simple deployment, and easy maintenance. However, existing vision-based methods rely on supervised learning for activity recognition, which is computationally intensive owing to the necessity of labeling large-scale training datasets. To address this problem, this paper describes a vision-based method for automatically analyzing excavators' productivities in earthmoving tasks by adopting zero-shot learning for activity recognition. The proposed method can identify activities of general construction machines (e.g., excavators and loaders) without pre-training or fine-tuning. To verify the feasibility, the proposed method has been tested on videos recorded from real construction sites. The accuracy values for activity recognition and productivity evaluation are 86% and 87.8%, respectively.

1. Introduction

Heavy construction machines (e.g., excavators, trucks, and loaders) play crucial roles in earthworks for intensively transporting soils and other materials at construction sites [1]. By continuously monitoring earthmoving operations, project managers can acquire machine activity and productivity information (e.g., number of cycles and idling time), which is essential for productivity analysis and resource allocation [2]. At present, earthmoving operations are typically monitored through manual approaches, which are labor-intensive and time-consuming. To overcome these limitations, information technologies have been introduced to automatically analyze earthmoving productivities by recognizing the construction machine activities [3].

Several researchers introduced sensor technologies such as the Global Positioning System (GPS) [4], ultra-wide band (UWB) [5], and inertial measurement units (IMUs) [6] to analyze the construction productivity. By processing the data retrieved from the sensors installed on construction machines for tracking their locations and movements in earthmoving projects, the activities and working cycles of the construction machines can be estimated. For example, Teizer et al. [7] used UWB for the automatic localization of earthmoving machines for productivity analysis. Moreover, Alshibani and Moselhi [8] estimated the

cost and time of earthmoving operations by using sampled GPS data. Notably, in these methods, sensors must be installed on either the construction sites or individual construction machines, which is not feasible or desirable for crowded sites and rented construction machines.

With the widespread installation of cameras on construction sites for surveillance, they have become ubiquitous devices in the construction industry worldwide. Compared with other sensors, cameras are less costly and can be easily deployed and maintained [9]. Therefore, many vision-based methods [10–12] have been developed for automatic machine monitoring and productivity analysis in construction tasks. The existing vision-based methods detect and track earthmoving machines (e.g., excavators) from videos and then recognize machine activities using deep learning methods. For example, Xiao and Kang [13] used three-dimensional (3D) convolutional neural networks (CNNs) for recognizing the activities of excavators to calculate the crew productivities in earthmoving projects. Roberts and Golparvar-Fard [14] proposed an end-to-end framework for calculating the productivities of earthmoving machines by integrating detection, tracking, and activity analyses. Overall, the development of a robust activity recognition method is the fundamental step for automatic productivity analysis in earthmoving projects

Most of the existing methods for vision-based productivity analyses

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rely on supervised learning to recognize the activities of construction machines, which involves two limitations: 1) The performance of the construction machine activity recognition depends on the quality and quantity of human-created datasets. However, the availability of high-quality visual data is limited in construction management. Moreover, frame-level annotation of the detailed activities of a machine object is labor-intensive. 2) The generalizability of the existing activity recognition methods is low because a model trained for a specific type of construction machine cannot be simply extended to other machines. For example, a deep learning model trained to recognize excavator activities cannot be applied for recognizing the activity of loaders.

To overcome the abovementioned limitations, the main objective of this research is to propose an automatic vision-based method for calculating the excavator earthmoving productivity based on zero-shot learning for activity recognition. Zero-shot learning is an artificial intelligence concept through which models pre-trained on daily life datasets can perform well in even unfamiliar scenarios (e.g., construction scenarios) without being fine-tuned on domain-specific datasets. In this research, a zero-shot learning method named contrastive language-image pre-training (CLIP) is used for activity recognition. Different from supervised vision-based methods, CLIP learns directly from raw text regarding images and reference the learned visual information through natural, thereby leveraging a broader source of supervision. In the proposed method, first, the excavators are detected and tracked to obtain their locations in video frames. The zero-shot learning classifier is applied to the tracked excavators to recognize their activities (e.g., excavating and loading) in each frame. According to the tracking and activity recognition results, the productivity of the excavators in earthmoving projects can be automatically calculated. The proposed method can reduce the time and cost required for developing annotated datasets for the vision-based monitoring of construction sites. Furthermore, the proposed method can be easily expanded to analyze the productivity of other construction machines (e.g., loaders) without pre-training or fine-tuning. To the best of the authors' knowledge, this study represents the first attempt at applying the zero-shot learning method in vision-based monitoring of construction sites.

2. Literature review

Continuous monitoring of construction sites by information technologies can facilitate construction management in terms of improved productivity, reduced safety risks, and accelerated project delivery. This section outlines the existing research related to the automatic monitoring of construction sites for activity recognition and productivity analyses. First, sensor-based monitoring methods are introduced, and then, vision-based monitoring methods are discussed. Finally, the research gaps are highlighted to clarify the motivation and objectives of this study.

2.1. Sensor-based monitoring methods

Sensor-based methods monitor construction sites by analyzing the position and movement data of construction machines collected from sensors. These methods can be categorized as real-time location systems (RTLS) methods [5,7,15,16], kinematic methods [1,17], or audio methods [18,19]. RTLS methods combine the data of positioning sensors (e.g., GPS and UWB) with the geographical information of the site to estimate the activity and productivity of a certain construction machine. For example, Montaser et al. [20] attached GPS sensors on trucks to obtain their trajectories. Subsequently, the hauling and returning activities were identified for calculating the truck productivities in earthmoving tasks. Vahdatikhaki and Hammad [21] used UWB to estimate the activities of excavators and trucks by dividing construction site into excavation, hauling, loading, and dumping regions. Notably, RTLS methods cannot accurately estimate the earthmoving productivities because the positioning information alone cannot be used to confirm

whether the machine is performing excavations in the working zone.

To solve this problem, kinematic sensors such as accelerometers and IMUs have been adopted for automatic productivity analyses in construction sites. Akhavian and Behzadan [22] attached 3D orientation sensors on the bed of a truck and boom of a loader. The activities (e.g., boom raising) were recognized by analyzing the change in the angles. Mathur et al. [23] mounted IMU sensors inside an excavator cabin to capture the vibration signals of the engine. Subsequently, the frequency features of the signals were extracted and classified to determine the cycle time of the excavator. Similar to RTLS methods, kinematic methods necessitate the installation of sensors on construction machines, which is not economically feasible in large-scale projects.

Audio methods recognize machine activities by classifying the features of audio signals [24]. For example, Cheng et al. [18] developed a framework to recognize machine activities by installing microphones on excavators, dozers, and loaders and analyzing the audio signals with the support vector machine (SVM) classifier. The authors later extended the framework by fine-tuning the software and hardware configurations [19]. Notably, audio methods may not be effective in noisy and crowded construction sites. Therefore, although sensor-based methods can enable the automatic monitoring of construction sites in a reliable manner, the installation, deployment, and maintenance of sensors are challenging, especially in large-scale projects.

2.2. Vision-based methods in the construction domain

Compared with sensor data, video footage is more accessible, given the widespread use of cameras in construction sites. Furthermore, cameras are less expensive and can be more easily deployed than sensors (e.g., laser scanners and GPS). With the rapid development of computer vision technologies, the performance of vision-based methods in construction management contexts has significantly improved. The existing vision-based methods for monitoring earthmoving operations can be divided into feature-based, tracking-based, and spatial-temporal CNN-based methods.

Feature-based methods extract image features from consecutive construction videos and encode them to fixed-length vectors to classify the activities of construction machines for productivity analyses. Gong et al. [25] used the histogram of optical flow and histogram of oriented gradients (HOG) methods to extract features from earthmoving videos and a Bayesian network to recognize the excavator activity for productivity estimation. Similarly, Golparvar-Fard et al. [12] used the 3D HOG technique for feature extraction and the SVM framework as the classifier to recognize the digging, swinging, hauling, and dumping activities of excavators in earthmoving projects. Notably, feature-based methods are not adequately accurate owing to their sensitivity to illumination condition variations in construction videos.

Tracking-based methods track construction machines from videos to obtain their trajectories for activity recognition and productivity analyses. Because these methods explore the contextual information in construction videos, they are more robust than feature-based methods. For instance, Bügler et al. [26] used the Gaussian mixture model to track excavators and trucks in earthmoving videos. If the pixel distance between an excavator and a truck was less than a pre-defined threshold, the excavator activity was recognized as loading, and the productivities were automatically calculated. Kim et al. [27] applied a HOG-based detector to obtain the positions of excavators and trucks. Subsequently, the working and idling activities were recognized based on the change in the centroid coordinates of the machine objects in continuous frames. Notably, in this method, the thresholds must be pre-defined in a case-by-case manner for recognizing machine activities, which is inconvenient in practice.

Recently, spatial-temporal CNN-based methods have achieved significant success in activity recognition and productivity analysis tasks in the construction automation field. Spatial-temporal CNNs are end-to-end deep neural networks that learn both spatial and temporal

information from continuous video frames. Compared with other methods, spatial-temporal CNNs are more accurate when trained with human-created activity datasets. Kim and Chi [11] created a 10-layer neural network that combined a CNN and long short-term memory network (LSTM) to recognize the digging, dumping, and hauling activities of excavators. Recognition accuracies of 79.8%, 90.9%, and 93.8% were obtained for digging, dumping, and hauling, respectively. Moreover, Chen et al. [28] integrated a 3D CNN to recognize the digging, swinging, and loading activities of excavators and calculate the earthmoving productivities. Accuracies of 87.6% and 83% were achieved in activity recognition and productivity analysis tasks, respectively. Although spatial-temporal CNN-based methods are highly accurate, the development of training datasets is expensive and time-consuming.

2.3. Research gaps and objectives

This discussion highlights that vision-based monitoring is a promising solution for automatically monitoring earthmoving operations. Spatial-temporal CNN-based methods can conduct end-to-end activity recognition with a high accuracy (i.e., over 80%), thereby outperforming other types of vision-based methods. This subsection summarizes the existing research gaps and objectives of this study.

2.3.1. Research gaps

Vision-based methods encounter the following two limitations in analyzing the earthmoving productivity:

1. The development of datasets for training spatial-temporal CNN-based methods is time-consuming and labor-intensive. Researchers must manually collect construction videos, edit video clips, and annotate clips at the frame-level. According to the authors' experience [13,28,29], the development of a dataset containing 100 video clips requires over 500 h of manual work over a period of three months. Owing to the difficulties in dataset development, the sizes of the existing datasets are inadequate (e.g., within 200 clips), which limits the performance of spatial-temporal CNNs in construction tasks.
2. The generalizability of the existing vision-based methods is low. In other words, a deep learning model trained for a certain type of construction machine cannot be applied to other machines. For example, the vision-based methods for monitoring the excavators' productivity cannot be directly applied for estimating the loaders' productivity. To overcome this problem, researchers must develop a new video dataset for loader activities and re-train the activity recognition model, which is time-consuming and labor-intensive. Thus, the low generalizability reduces the flexibility of vision-based methods in earthmoving projects.

2.3.2. Research objectives

To fill the abovementioned gaps, this research is aimed at establishing a novel method to calculate the excavator productivity in earthmoving tasks based on zero-shot learning for activity recognition. The two key objectives of this study are as follows:

1. Perform zero-shot learning for recognizing the activities of excavators in earthmoving operations and validate the generalizability of zero-shot learning methods applied in construction scenarios. Unlike the existing activity recognition methods, zero-shot learning does not require construction-specific training datasets, and thus, the model can be easily extended to other types of machines.
2. Develop a framework for analyzing excavator productivities in earthmoving projects by integrating object detection, object tracking, and zero-shot learning for activity recognition. The proposed framework is expected to automatically determine the productivity and idling time of excavators with a satisfactory accuracy.

The findings can help demonstrate the feasibility of zero-shot learning in earthmoving productivity analyses.

3. Methodology

A novel method that integrates object detection, object tracking, activity recognition, and productivity evaluation is developed for automatically analyzing the productivity in earthmoving operations. Fig. 1 shows the framework of the proposed method. When a video is input to the proposed method, the object detection module recognizes the excavator objects from each frame by generating the corresponding pixel locations of these objects. These detection results are input to the object tracking module for associating the excavator objects across frames, and a unique identification (ID) number is assigned to each excavator object that appears in the video. The tracked excavator areas are cropped and fed to the activity recognition module to identify the excavators' activities from one of the following categories: loading, excavating, and idling. Activity recognition is performed based on zero-shot learning that does not require construction-specific datasets for training. The activity recognition results are input to the productivity evaluation module to estimate the productivity rate and idling time of the excavators. Typically, the productivity is calculated based on the results of the three modules, and thus, the calculation accuracy depends on the detection, tracking, and activity recognition results. This process can be used to automatically determine the crew productivity in earthmoving operations. The following subsections describe the object detection, object tracking, activity recognition, and productivity evaluation modules.

3.1. Object detection

The object detection module detects the excavator objects from the input video sequences. To this end, it is necessary to use a robust object detection method that produces precise bounding boxes of excavator objects. In this study, the deep learning object detector You Look Only Once (YOLO-v5) [30] is used for detecting excavators owing to its reliable performance and high processing speed. The reliability of YOLO-v5 in detecting construction-related objects has been previously demonstrated [31–33].

As shown in Fig. 2, YOLO-v5 has an end-to-end detection architecture that incorporated a cross-stage partial network (CSPNet), path aggregation network (PANet), and YOLO layer. The CSPNet is the backbone for YOLO-v5 and helps eliminate the problems of repeated gradient information in the training stage. The PANet is a feature pyramid network with an enhanced bottom-up path that helps improve the detection performance, especially for small-scale objects. The YOLO layer generates three types of feature maps (18×18 , 36×36 , 72×72) for multiple-scale detection to facilitate detection. Notably, in this study, the YOLO-v5x model is used, given that it has achieved a mean average precision (mAP) of 68.9% on the COCO dataset [34] and a processing speed of 82.6 frames per second (fps) on the Amazon Web Services p3.2 large instance.

An annotated dataset is required for training deep learning object detectors for domain tasks. In this study, the Alberta Construction Image Dataset (ACID [29]) is used for training the YOLO-v5 construction machine detector. The ACID contains 10,000 annotated construction images of ten types of machines: excavators, compactors, dozers, graders, dump trucks, concrete mixer trucks, wheel loaders, backhoe loaders, tower cranes, and mobile cranes. These images have been collected from online sources (e.g., YouTube, Google Image, and Naver) and onsite sources (e.g., smartphones, drones, and stationary cameras), thereby ensuring data diversity. The YOLO-v5 construction machine detector trained on ACID can detect excavator objects from the images input to the object tracking module.

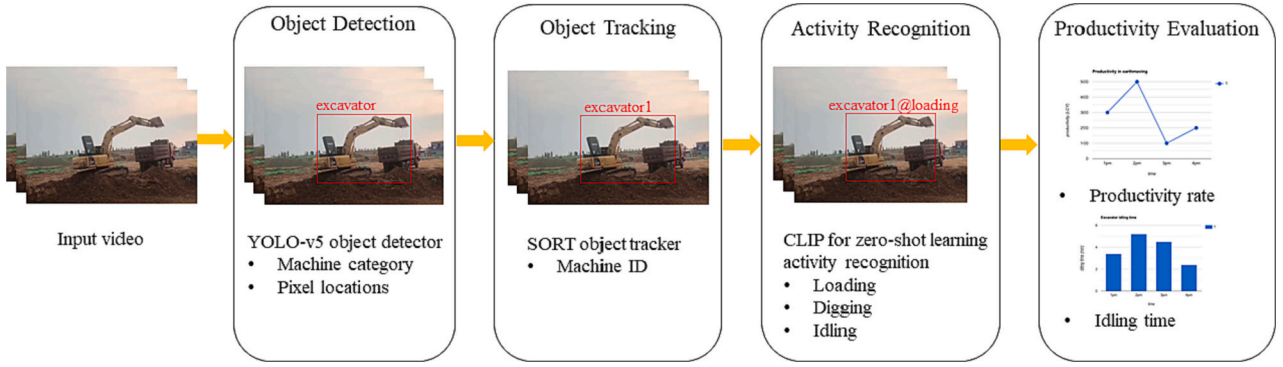


Fig. 1. Framework of the proposed method.

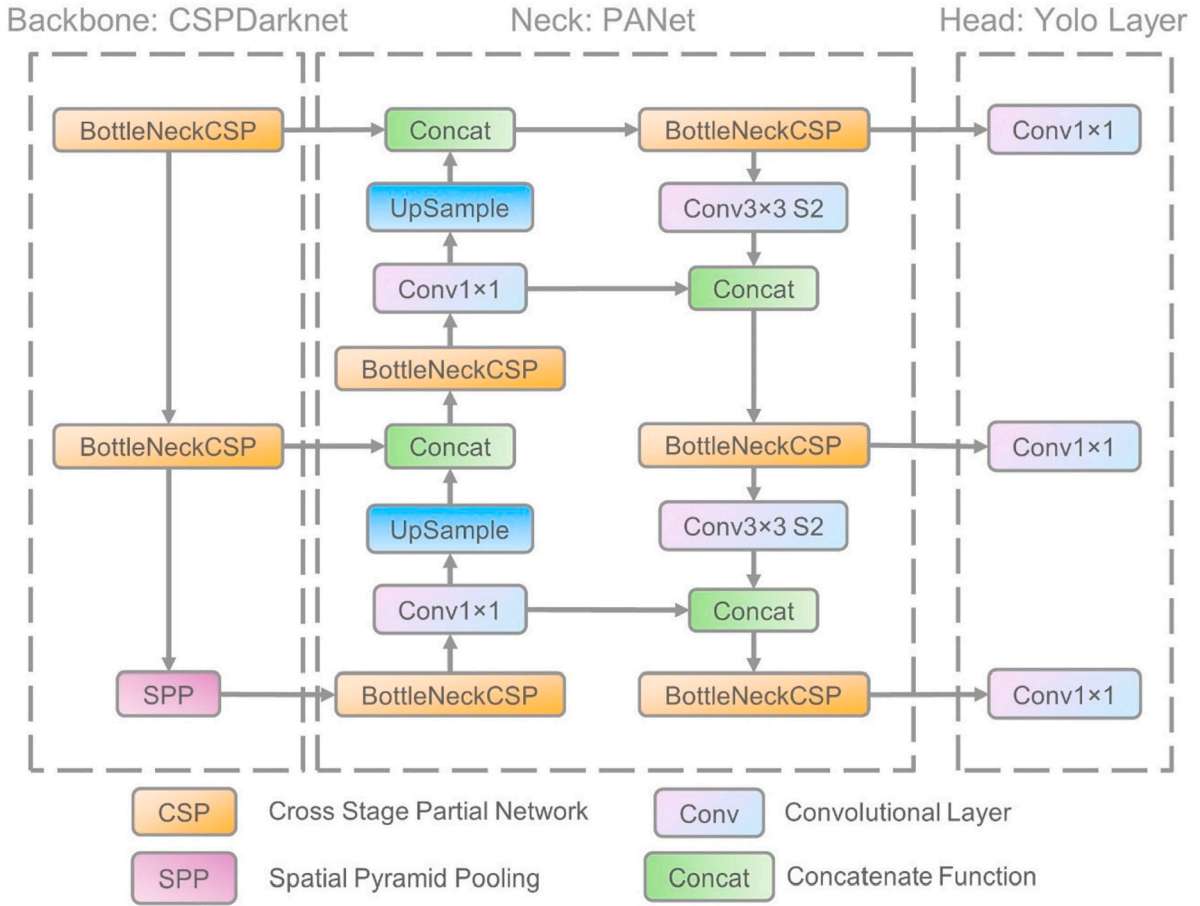


Fig. 2. YOLO-v5 network architecture (image reprint from [35]).

3.2. Object tracking

The object tracking module retrieves the excavator trajectories by associating the detection results across frames. The tracking module assigns a unique and consistent ID for each detected object that appears in continuous frames. The activity recognition module can analyze the changes in the excavator objects in continuous frames to classify their activities. In the proposed method, the simple online and real-time tracking (SORT) method [36] is used for tracking excavators as its tracking performance has been noted to be satisfactory on the MOT16 dataset [37]. Furthermore, SORT relies on only the pixel position information of objects, and thus, it enables rapid tracking (260 fps) for even multiple objects.

The SORT method consists of three steps: Kalman filtering prediction

[38], data association, and linear assignment. In the first step, tracking bounding boxes are generated on a certain frame according to the trajectory information in previous frames. Moreover, YOLO-v5 produces detection bounding boxes for this frame. In this second step, a matrix is built by calculating the intersection over union (IoU) between each pair of detection and tracking bound boxes. In this manner, the tracking problem is converted to a linear assignment problem. Finally, the Hungarian algorithm [39] is applied to solve the IoU matrix to produce the tracking results for this certain frame. The tracking regions of each excavator object in continuous frames are input to the activity recognition module.

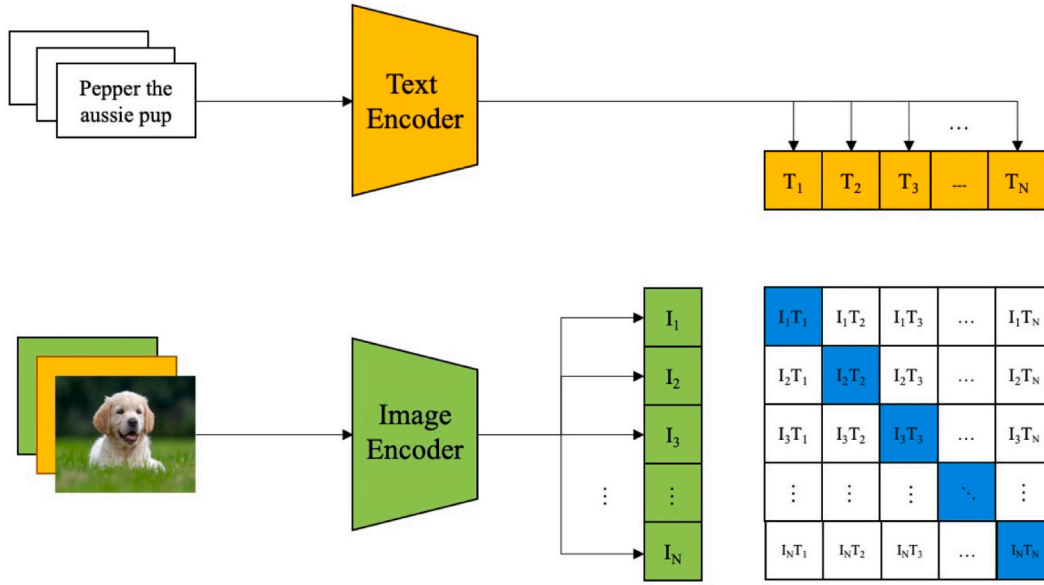
3.3. Activity recognition

The activity recognition module classifies the activities of the tracked excavator objects in each frame into either loading, digging, or idling. This module involves two steps, the initial inputs of which are the object tracking results: 1) Process all the frames in the video to identify whether the excavator is idling; and 2) input all non-idling frames into the zero-shot learning neural networks to identify whether the excavator is loading or digging.

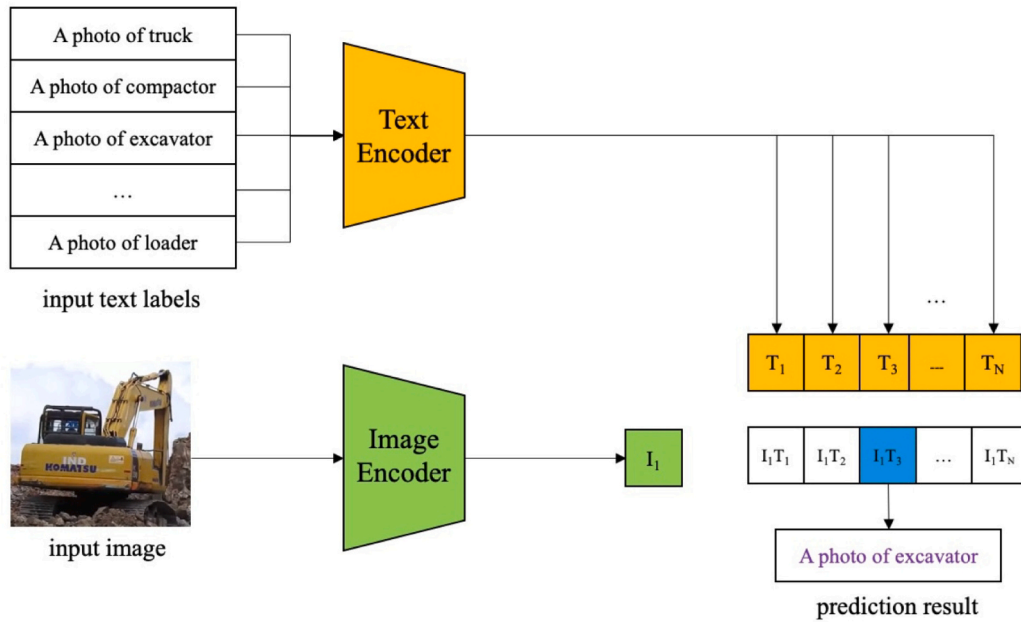
When an excavator is idling, the engine is not operating, and the corresponding tracking boxes change only slightly. Therefore, the pixel

locations of the excavators can be used for identifying whether the excavator is idling, as reported previously [13,14,28]. In this study, the centroid point of the tracked excavator object at frame i is denoted as (x_i, y_i) . The change in the centroid point (d_i) in frame i can be calculated as the Euclidean distance of the excavator between frames i and $i - fr$ (Eq. 1), and the average centroid point change (ad_i) in fr continuous frames can be calculated using Eq. 2, where fr is the frame rate of the video. If ad_i is smaller than the threshold value d' , the excavator is identified to be idling at frame i . Otherwise, this frame is identified to be non-idling.

$$d_i = \sqrt{(x_i - x_{i-fr})^2 + (y_i - y_{i-fr})^2} \quad (1)$$



(a) Contrastive pre-training



(b) Zero-shot prediction

Fig. 3. CLIP methodology.

$$ad_i = \frac{1}{fr} \sum_{n \in (i-fr, i]} d_n \quad (2)$$

Subsequently, the non-idling frames are input to the zero-shot learning framework (CLIP) [40] for recognizing loading and digging activities. Using the fed image and several text labels, the CLIP can efficiently predict the text label that corresponds to the image, which can then be used for various tasks such as object classification, object detection, activity recognition, and optical character recognition. As shown in Fig. 3, CLIP involves two steps: contrastive pre-training and zero-shot prediction.

In pre-training, given a batch of N (image, text) pairs, the image encoder ResNet50 [41] and text encoder Vision Transformer (ViT) [42] retrieve image features I_N and text features T_N from each pair of image and text, respectively. The text encoder transfers the features of the batch into vectors $[T_1, T_2, \dots, T_n]$ with (N, d_t) dimensions, and the images are transferred to the vectors $[I_1, I_2, \dots, I_n]$ with (N, d_i) dimensions. The text vector is correlated with the vision vector, for example, T_1 is correlated with I_1 . These N correlated pairs are termed positive samples, and the vectors that are not related are termed negative samples. The number of negative samples is $(N^2 - N)$. The positive and negative samples can be used as labels for training. The text and image can be matched by calculating the cosine similarity between T_i and I_i . In the training stage, a multi-modal embedding space is introduced to jointly train the image encoder and text encoder by maximizing the cosine similarity of the image and text embeddings in each batch. In the zero-shot prediction step, one image and several texts are input to the trained image and text encoder, respectively. Subsequently, the image and text are transformed to vectors to calculate the cosine similarity. The text with the highest cosine similarity with the image is selected as the prediction result. Note that CLIP has been trained with 400 million (image, text) pairs collected from various publicly available sources on the Internet. To cover as broad a set of visual concepts as possible, a base query list (500,00 queries) that contains all the words occurring at least 100 times in the English version of Wikipedia was prepared. The (image, text) pairs whose text is included in the base query were added to the dataset, and approximately 20,000 (image, text) pairs were added per query. The total word count of the dataset is similar to that of the WebText dataset, which contains 8 million documents for 40 GB of text.

In this study, two texts, “the excavator is loading dirt to the truck” (representing loading) and “the excavator is digging dirt from the ground” (representing digging), are used as the text inputs for CLIP. The loading and digging activities can be identified by inputting all non-idling frames into the CLIP without fine-tuning. Digging and loading activities of the excavator are selected for this task because the excavator’s productivity depends on the number of buckets loaded by the excavator to the trucks. The intermediate activity between digging and loading, i.e., swinging, is ignored as its duration is too short to be recognized.

3.4. Productivity evaluation

The productivity evaluation module calculates the excavator’s productivity and idling time using the activity recognition results. Notably, the activity recognition results may contain systematic errors [28]. Therefore, we first correct the activity recognition results by applying

majority voting every fr frames to ensure that the excavator has the same activity at every second. Typically, fr is determined based on the operating speed of the excavator and fps of the video. In this study, fr is defined as 25 frames, equivalent to a 1 s video, because the excavators are operating continuously and cannot change their activities in a short period. Moreover, according to site surveillance video observations, each activity lasts at least 1 s during the operation. Fig. 4 shows an example of the correction of abnormal activity recognition results. The assumptions in this process are reasonable because the excavator movement is slow and consecutive in real earthmoving operations.

After correction, the idling time can be calculated as the number of idling frames divided by the video frame rate fr (Eq. 3). The duration of one earthmoving cycle can be estimated as the number of frames between two adjacent loading activities divided by fr , and the number of earthmoving cycles is equal to the number of loading activities that appear in the video. Therefore, the excavator productivity can be calculated using Eq. 4, where the bucket payload per cycle is decided by the manufacturer, and LCY refers to the loose cubic yard.

$$idling\ time = \frac{number\ of\ idling\ frames}{fr} \quad (3)$$

$$productivity = \frac{number\ of\ cycles}{time\ (h)} \times \frac{bucket\ payload}{cycle}\ (LCY) \quad (4)$$

4. Implementation, experiment, and result analysis

The proposed method is implemented using Python 3.7 with Opencv 4.5 for video input/output (I/O). YOLO-v5 and CLIP are implemented using the Pytorch library, and SORT is implemented using the filterpy and numpy libraries. All the modules are integrated using a Python wrapper, and the proposed method is tested in an Ubuntu 20.04 64-bit system environment with acceleration via CUDA Toolkit 11.5 and cuDNN 8.3.3. The hardware setup involved a computer with two NVIDIA GeForce RTX 3070 GPUs with 8 GB graphic memory, an Intel Core i7-10700KF@3.80 GHz CPU, and two 32 GB memory cards.

Two experiments are conducted to validate the proposed method. The first experiment is aimed at evaluating the feasibility of the zero-shot activity recognition module in construction scenarios. A test dataset containing 50 short video clips of excavators in earthmoving operations is collected. The proposed activity recognition module is used to identify the excavator activities in the test dataset. The second experiment is aimed at validating the automated productivity analysis capability of the proposed method for long earthmoving videos. In this manner, the overall performance of the proposed object detection, tracking, and activity recognition modules can be evaluated. The following subsections describe the experimental setup and results.

4.1. Activity recognition experiment and results

The first experiment is aimed at evaluating the performance of the zero-shot learning based activity recognition module (CLIP) in earthmoving operations, which is important because the performance of activity recognition directly affects the accuracy of the productivity calculation. An activity recognition dataset of excavators is established with reference to existing studies [11,28]. Fig. 5 shows example video frames from the dataset that contains 50 short video clips of the

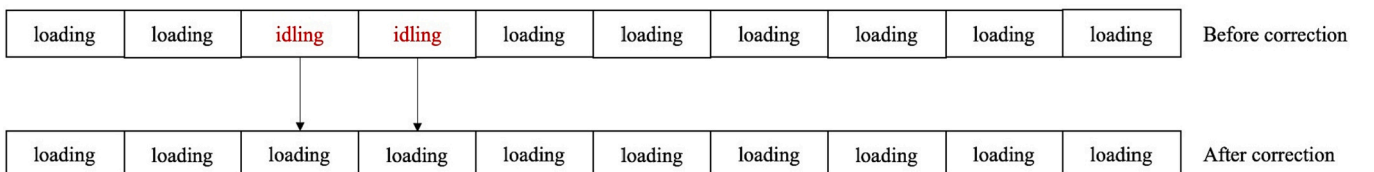
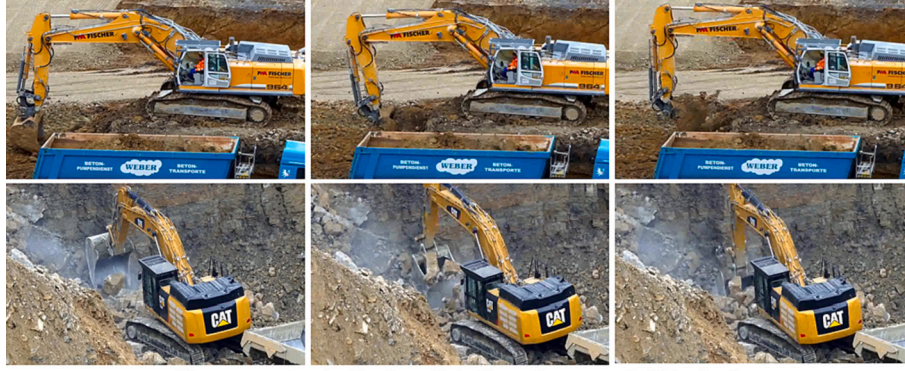
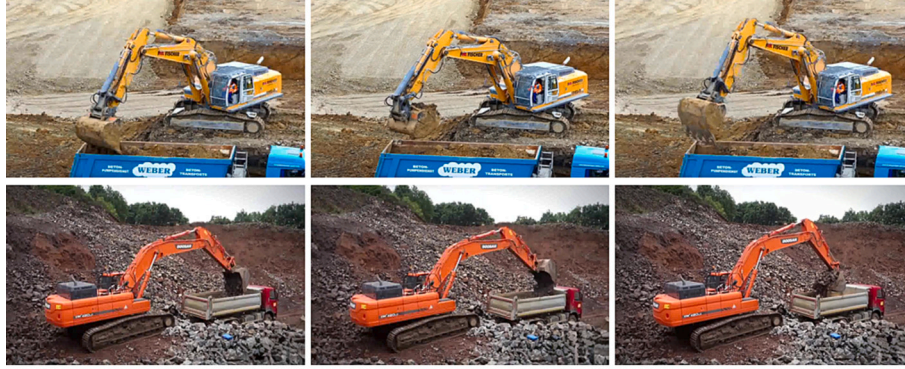


Fig. 4. Abnormal correction for activity recognition results.



The excavator is digging dirt from the ground (Digging)



The excavator is loading dirt to the truck (Loading)

Fig. 5. Examples of excavator activity recognition dataset.

excavator performing two types of activities (i.e., digging and loading) in earthmoving operations. To validate the performance of CLIP in different construction scenarios, the test videos are collected from seven construction sites under different viewpoints, colors, and light conditions of the excavators. Table 1 presents the statistical information of the test dataset. The duration of the videos is between 3 and 9 s, and all videos have a framerate of 25 fps. In summary, 6750 video frames with a resolution of 1280×720 pixels are used in this experiment.

As described in Section 3, CLIP does not require additional fine-tuning with domain-specific datasets because training is performed using a general and adequately common large-scale dataset. Two labels are input to the CLIP: “the excavator is digging dirt from the ground” (representing digging) and “the excavator is loading dirt to the truck” (representing loading). For each testing video, CLIP processes all frames and identifies if the excavator in this frame is digging or loading. Subsequently, major voting is applied to smooth the recognition results. If two-thirds of the frames in a video correspond to a certain activity, the video clip is annotated as this activity; otherwise, recognition failure is considered to have occurred.

The experimental results are summarized in Table 2. Three indicators, i.e., accuracy, precision (Eq. 5), and recall (Eq. 6) are applied to validate the performance of the CLIP.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

where TP is the number of true positives, FP is the number of false positives, and FN is the number of false negatives. More details regarding the concepts of accuracy, precision, and recall can be found in the work of Kim et al. [27]. In the experiments, 22 digging video clips are successfully recognized with an accuracy of 88%. Moreover, 21

Table 1
Statistical information of excavator test dataset.

Number	Digging		Loading	
	Video time (s)	Frame number	Video time (s)	Frame number
1	5	125	4	100
2	6	150	3	75
3	4	100	4	100
4	5	125	4	100
5	6	150	4	100
6	5	125	4	100
7	3	75	5	125
8	9	225	4	100
9	8	200	3	75
10	6	150	4	100
11	7	175	4	100
12	9	225	4	100
13	5	125	2	50
14	5	125	6	150
15	9	225	7	175
16	3	75	4	100
17	3	75	5	125
18	7	175	5	125
19	7	175	6	150
20	6	150	6	150
21	5	125	5	125
22	6	150	7	175
23	7	175	9	225
24	6	150	6	150
25	6	150	7	175
Average	5.92	148	4.88	122
Total time (s)	148	—	122	—
Total frames (s)	—	3700	—	3050

Table 2
Experimental results for activity recognition.

	Number of activities	Correctly detected activities	Precision (%)	Recall (%)	Accuracy (%)
Digging	25	22	95.70	88.00	
Loading	25	21	100.00	84.00	
Average/total	50	43	97.80	86.00	86.00

loading video clips are correctly identified with an accuracy of 84%. Overall, the proposed activity recognition method achieves an accuracy of 86%, comparable with that of the existing spatial-temporal CNN methods. For example, an activity recognition accuracy of 87% was achieved in a previous study [28] based on the same evaluation method. In other words, the proposed zero-shot learning based activity recognition method achieves state-of-the-art accuracy with reduced effort of annotating construction-specific datasets and reduced use of computational resources for fine-tuning the deep learning model. Fig. 6 shows the results of activity recognition of excavators in earthmoving work. The types of activities are indicated in each video frame by the pre-defined text labels.

4.2. Productivity analysis and results

The objective of this study is to establish a novel productivity analysis method for earthmoving construction. Therefore, the capability of the proposed method in long earthmoving videos must be evaluated. In this experiment, a construction video (1 h 3 s) is captured from a real construction site in Montreal City, Quebec, Canada to validate the performance of the proposed method. Fig. 7 shows example frames of the testing video. The framerate of the video is 30 fps, and the resolution is 1920 × 1080. In the video, a Deere 300G LC excavator (Excavator 1) (bucket payload of 1.7 LCY) is digging dirt and loading it to dump trucks, and another excavator (Excavator 2) is digging and idling. Both excavators are detected and tracked, and their activities are recognized. Because Excavator 2 does not perform loading, the productivity of only Excavator 1 is evaluated, and the threshold d' is set as three pixels for the idling analysis. The sensitivity analysis results for selecting the thresholds for the majority voting algorithm and idling recognition are presented in Appendix A.

Table 3 summarizes the experimental results. Excavator 1 processes 41 buckets and has an idling period of 40 min 39 s. Therefore, the

ground truth productivity of Excavator 1 is 69.64 LCY/h. The proposed method successfully identifies 36 buckets (productivity of 61.14 LCY/h), and the accuracy is 87.8%. The identified idling time is 43 min 32 s. The test results demonstrate that the proposed framework can accurately calculate the productivity of excavator from construction surveillance videos. Fig. 8 presents example images of the productivity analysis results, with the activity recognition results presented as pre-defined text labels.

5. Discussion

The experimental results show that the proposed method can accurately recognize the activities of the excavators in earthmoving works. This section discusses the implications of these results and limitations of this study.

5.1. Detection performance of YOLO-v5

The proposed method integrates object detection, tracking, and activity analysis to automatically calculate the productivity in earthmoving operations. Therefore, the backbone object detector is a core component of the proposed method, and its effectiveness influences the accuracy of productivity calculation results. In this study, the YOLO-v5 model is used for object detection, which can detect ten types of construction machines after training over ACID. The following experiment is conducted to validate the backbone detector performance: YOLO-v5 is trained on the ACID training set (80%), and the trained models are tested over the ACID validation set (20%). The performance is evaluated in terms of the mAP (Eqs. 7 and 8).

$$AP = \frac{1}{11} \sum_{recall \in \{0.0, 0.1, 0.2, \dots, 1\}} Precision@recall \quad (7)$$

$$mAP = \frac{1}{N} \sum AP \quad (8)$$

where precision and recall are determined using Eqs. 5 and 6, respectively. N refers to the number of classes in the object detection process. The detection bounding box is considered to be correct when the IoU between the detected and ground truth is larger than 0.5.

Table 4 summarizes the performance of YOLO-v5 and two other state-of-the-art detection methods (YOLO-v3 [43] and Faster-RCNN [44]) on the ACID dataset for detecting construction machines. YOLO-v5 achieves an mAP of 92.4%, which is higher than those of the other detection methods. Moreover, YOLO-v5 can effectively detect machines

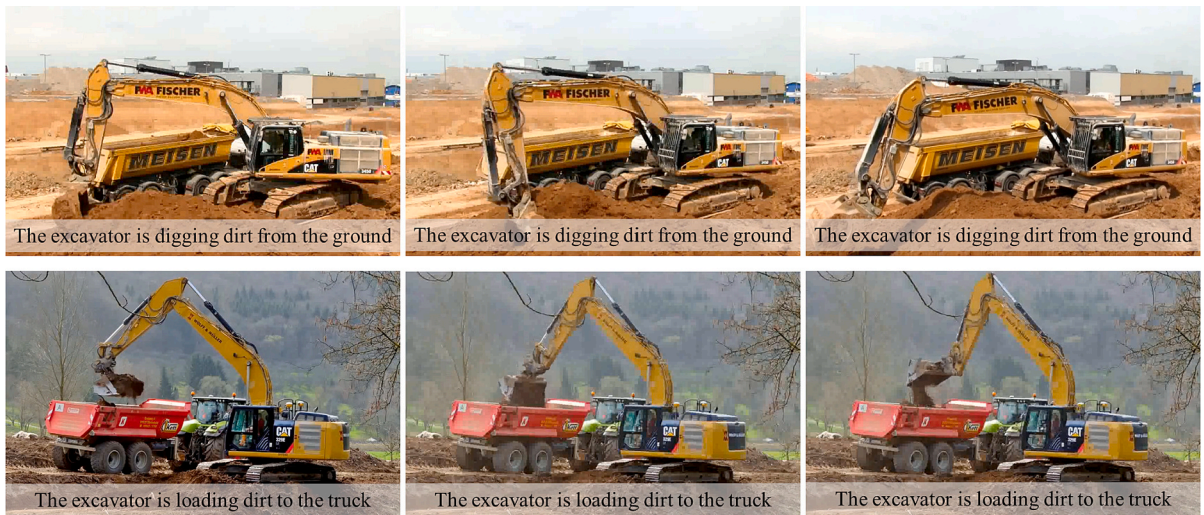


Fig. 6. Examples of excavator activity recognition test results.



Fig. 7. Example frames of test video for productivity evaluation.

Table 3

Productivity calculation results.

	Number of buckets	Idling time	Productivity (LCY/h)
Ground truth	41	40 min 39 s	69.64
Proposed method	36	43 min 32 s	61.14
Accuracy	87.8%	93%	87.8%

that are widely employed in earthmoving works (e.g., 87.5% AP for dump trucks and 94.2% AP for excavators). Therefore, YOLO-v5 is a robust module for the proposed method for vision-based earthmoving productivity analysis.

5.2. Generalizability of the proposed method

Because the activity module in the proposed method does not require fine-tuning, it is general enough to be extended for the productivity analysis of other construction machines used in earthmoving work. An experiment is conducted to evaluate the performance of the proposed activity recognition module for loaders. A testing dataset containing video clips of loaders in earthmoving work is generated. Fig. 9 shows example images of this test dataset. The dataset has 40 video clips of two activities (i.e., digging and loading) of loaders, collected from 14 construction sites. The framerate and resolution of each video clip are 25 fps and 1280×720 pixels. Table 5 summarizes the dataset information.

The same experimental setup as that described in Section 4.1 is used. “The loader is loading dirt to the truck” and “the loader is digging dirt from the ground” are input to the activity recognition module as pre-defined text labels representing the loading and digging activities,

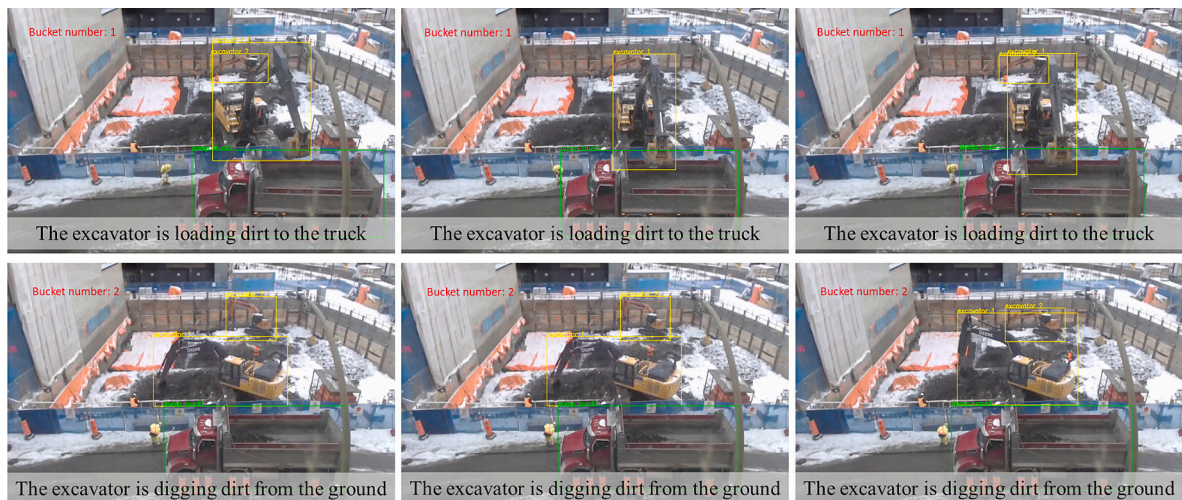


Fig. 8. Examples of productivity analysis results.

Table 4

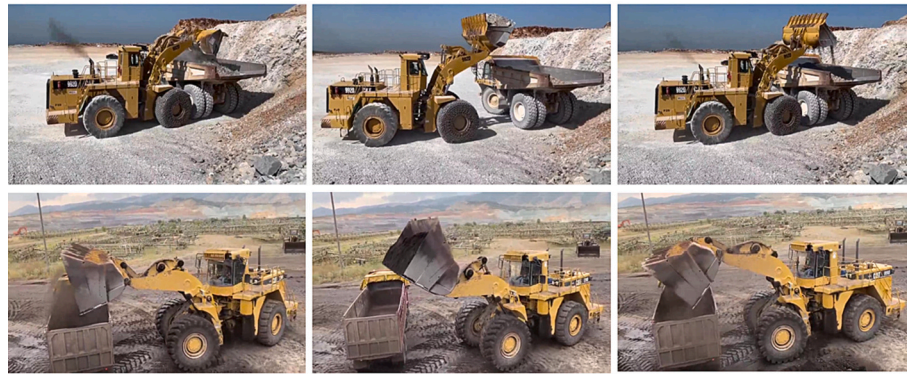
Performance analysis of object detection methods in construction.

Detection method	AP (%)										mAP (%)
	Excavator	Compactor	Dozer	Grader	Dump truck	Concrete mixer truck	Wheel loader	Backhoe loader	Tower crane	Mobile crane	
YOLO-v5*	94.2	94.2	94.5	98.9	87.5	96.4	92.9	96.1	80.0	89.3	92.4
YOLO-v3	93.5	94.8	89.9	95.1	83.3	94.9	84.6	95.6	62.0	84.5	87.8
Faster-RCNN	92.5	92.3	95.6	98.7	81.5	92.6	90.6	95.9	64.4	88.4	89.2

* Detection method adopted in the present study.



The loader is digging dirts from the ground (Digging)



The loader is loading dirts to the truck (Loading)

Fig. 9. Examples of loader activity recognition dataset.

respectively. Fig. 10 shows example results for the activity recognition of loaders in earthmoving works. The results are summarized in Table 6. The proposed module correctly identifies 17 digging clips and 16 loading clips, achieving an average accuracy of 82.5% in recognizing the loader activities in earthmoving work.

The proposed zero-shot learning based method achieved an accuracy of more than 80% in recognizing the activities of both excavators and loaders, which demonstrates the generalizability of zero-shot learning in activity recognition in construction scenarios. Compared with supervised learning methods, zero-shot learning methods can be easily extended to other types of construction activities, thereby reducing the effort required for training dataset development and model fine-tuning. The proposed productivity analysis method that incorporates zero-shot learning can be conveniently applied to evaluate the earthmoving productivities of multiple types of construction machines. Moreover, the zero-shot learning methods can be applied to other vision-based applications in construction management, e.g., for safety control and productivity analysis, by recognizing worker activities owing to its high generalizability.

5.3. Comparison with other methods

To demonstrate the novelty of the proposed method, it is compared with existing methods for vision-based earthmoving operation monitoring. Table 7 presents the comparison results in terms of the publication year, method description, target machines, training dataset, activity recognition results, and productivity analysis results. The proposed method achieves a high accuracy (over 80%) and does not require an extensive training process. In contrast, most of the existing methods require manual effort for collecting and annotating large-scale datasets for training (typically, a few hundred video clips) the activity recognition model. Specifically, the development of such video-based machine activity recognition datasets requires the trimming of video clips and annotation of the video clips at the frame level, which are expensive and time-consuming and may involve copyright and privacy issues. Furthermore, the proposed method exhibits reliable performance (accuracy of 87.8%) in the productivity analysis based on an earthmoving video with a duration of 1 h 3 s, which demonstrates the feasibility of using it for monitoring actual construction sites. The speed of activity recognition of the proposed method is approximately 20 fps in the current configuration, which is considerably higher than that of spatial-temporal CNN methods (10 fps) [45] in the same environment.

Table 5
Statistical information of test dataset for loader activity recognition.

Video number	Digging		Loading	
	Video time (s)	Frame number	Video time (s)	Frame number
1	23	575	9	225
2	6	150	9	225
3	5	125	8	200
4	4	100	8	200
5	4	100	8	200
6	2	50	7	175
7	2	50	6	150
8	5	125	7	175
9	4	100	4	100
10	3	75	9	225
11	5	125	8	200
12	9	225	7	175
13	11	275	7	175
14	7	175	5	125
15	5	125	4	100
16	6	150	7	175
17	20	500	6	150
18	23	575	7	175
19	15	375	8	200
20	5	125	6	150
Average	8.2	205	7	175
Total time (s)	164	—	140	—
Total frames	—	4100	—	3500

5.4. Limitations of the proposed method

The proposed model involves the following three limitations: (1) In this given configuration, the idling time is analyzed using an intrinsic method, and the threshold must be manually set in different construction scenarios. This aspect is a general limitation of vision-based monitoring and can be overcome by using sensors for idling analysis. (2) The accuracy of the proposed method in productivity analyses is 87.8%. Although this value acceptable, it is slightly lower than those of the existing methods based on supervised learning owing to the limitations of CLIP. The CLIP performance can be enhanced by introducing the attention mechanism in its text encoder. (3) The productivity is calculated based on the number of buckets of the excavator. However, the amount of soil in the bucket may not be equal to the standard payload and may vary because of other reasons, such as the operators' skill. This is also an inherent limitation of vision-based methods and can be overcome by installing pressure sensors on the buckets. (4) In this study, the zero-shot learning based activity recognition was evaluated only for excavators and loaders, and the productivity analysis was

performed only for excavators. The feasibility of the proposed method for other earthmoving machines such as bulldozers, backhoes, and road rollers was not evaluated owing to the limited resources for testing dataset development. More comprehensive experiments must be conducted to validate the proposed method.

6. Conclusions

This paper presented a vision-based monitoring method for productivity analysis. The proposed method contains four major modules: object detection, object tracking, activity recognition, and productivity evaluation. A zero-shot learning method (CLIP) is introduced to recognize machine activities in earthmoving tasks. The proposed method can recognize the activities of an excavator and a loader with average accuracies of 86% and 82.5%, respectively. The accuracy is 87.8% when testing is performed on a 1 h 3 s earthmoving video captured from a real construction site.

The contributions of this study are three-fold: First, a framework that integrates vision-based construction machine detection, tracking, and activity recognition methods is developed to automatically determine the productivity of construction machines. Second, a database-free method, CLIP, is used to recognize the activities of multiple types of construction machine such as excavators and loaders. Third, the possibility of applying zero-shot-based methods for construction machine monitoring is validated. Owing to the generalizability of zero-shot-based methods, construction machine monitoring can be efficiently performed.

Future work can be focused on addressing the following issues. First, the idling identification module must be improved by integrating sensor-based methods. Second, the activity recognition capability for other types of construction machines (e.g., bulldozers, backhoes, and road rollers) and longer videos must be tested. Third, the accuracy of the proposed method can be enhanced by introducing attention and

Table 6
Test results for recognizing loader activities in earthmoving.

	Number of activities	Correctly detected activities	Precision (%)	Recall (%)	Accuracy (%)
Digging	20	17	89.50	85.00	
Loading	20	16	88.90	80.00	
Average/total	40	33	89.20	82.50	82.50

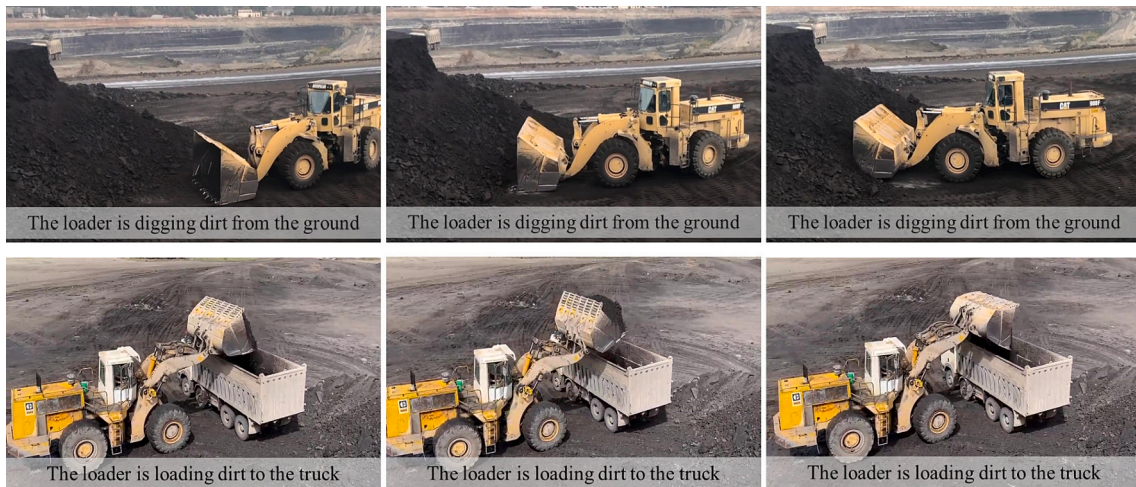


Fig. 10. Examples of loader activity recognition test results.

Table 7

Comparison of the proposed method and existing methods for vision-based productivity analysis in earthmoving operations.

Reference	Year	Method	Target machines	Training dataset for activity recognition	Activity recognition accuracy	Productivity analysis accuracy
Golparvar-Fard et al. [46]	2013	Histogram of gradients for feature extraction and support vector machine (SVM) for activity recognition	Excavator and dump truck	600 video clips	86.3% for excavator; 98.3% for dump truck	N/A
Kim et al. [47]	2018	Tracking-learning-detection (TLD) for machine detection and tracking and rule-based interaction analysis for activity recognition	Excavator and dump truck	11,513 frames from a 150 min video	89.4% for excavator; 93.8% for dump truck	N/A
Kim and Chi [11]	2019	Faster R-CNN for machine detection, TLD for machine tracking, and CNN-LSTM for activity recognition	Excavator	72,365 frames from a 121 min video	88.1%	N/A
Roberts and Golparvar-Fard [14]	2019	ResNetx101 for machine detection, FCNT tracker for machine tracking, and hidden Markov model with SVM for activity recognition	Excavator and dump truck	514 video clips	86.8% for excavator; 88.5% for dump truck	N/A
Chen et al. [28]	2020	YoloV3 for machine detection, Deep SORT for machine tracking, and 3D CNN for activity recognition	Excavator	351 video clips	87.6%	83% for a 60.2 min earthmoving video
Proposed	2022	YoloV5 for machine detection, SORT for machine tracking, and CLIP for activity recognition	Excavator and loader	Not needed	86% for excavator 82.5% for loader	87.8% for a 60.03 min earthmoving video

transformer mechanisms for text and image encoders, respectively.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Experiments for threshold selection

Threshold selection for majority voting

In the majority voting process, the activity recognition results of all frames are fed into the majority voting algorithm. Then, the algorithm is implemented with a threshold of 2/3 for activity recognition. In other words, the activity recognized in two-thirds of the frames is regarded as the activity of the video. In the existing studies, the majority voting method has been used for smoothing the activity recognition results. In general, if over half the frames correspond to the same activity, the video considered to correspond the major activity. When this strategy is applied in this study, an accuracy of approximately 100% is obtained. However, in several clips, the number of two activities is identical, which means that the activity recognition module is sensitive. To overcome this issue, an experiment is conducted to explore the use of different thresholds. Based on the results presented in Table A1, a threshold of 2/3 is selected.

Table A1

Sensitivity analysis for selecting threshold for majority voting.

Major voting threshold	1/2	2/3	5/6
Digging accuracy (%)	100	88	88
Loading accuracy (%)	96	84	84
Average accuracy (%)	98	86	86

Threshold selection for idling identification

The threshold for idling identification is selected based on the sensitivity analysis results presented in Table A2. When the threshold d' is larger than 3, the idling time becomes longer, more loading and digging are misidentified as idling, and the productivity accuracy decreases.

Table A2
Sensitivity analysis for selecting threshold for idling identification.

d' (pixels)	Recognized/ground truth number of bucket loadings	Productivity analysis accuracy (%)
1	30/41	73.2
2	32/41	78.0
3	36/41	87.8
4	35/41	85.4
5	35/41	85.4

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